

Web Technology



React Routing

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Front End Development Tasks

App state	Data definition, organization, and storage
User actions	Event handlers respond to user actions
Templates	Design and render HTML templates
Routing	Resolve URLs
Data fetching	Interact with server(s) through APIs and AJAX

React Router

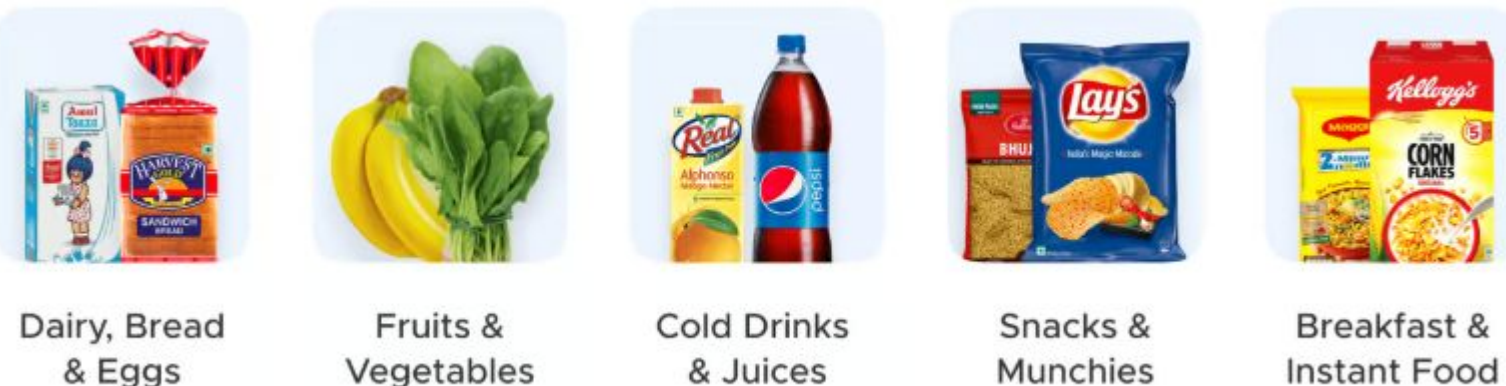
1. **Client-Side Routing**
2. **Routing?**
 - **Directing to Different Pages**

React Router

1. Client-Side Routing

2. Routing?

- Directing to Different Pages



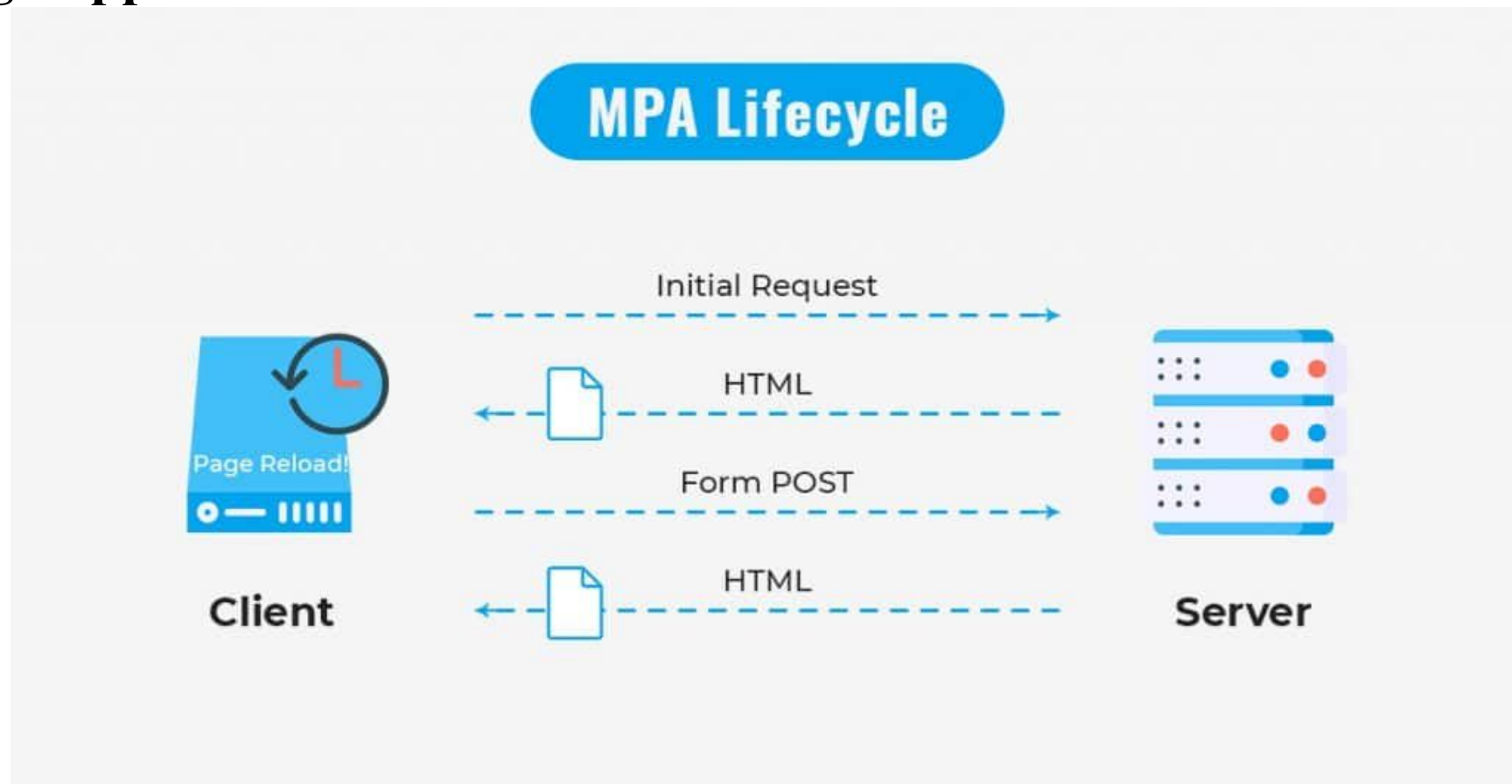
React Router

1. **Client-Side Routing**
2. **Routing?**
 - **Directing to Different Pages**
 - **Internal Routing**
 - **External Routing**

React Router is primarily for internal routing within a React application.

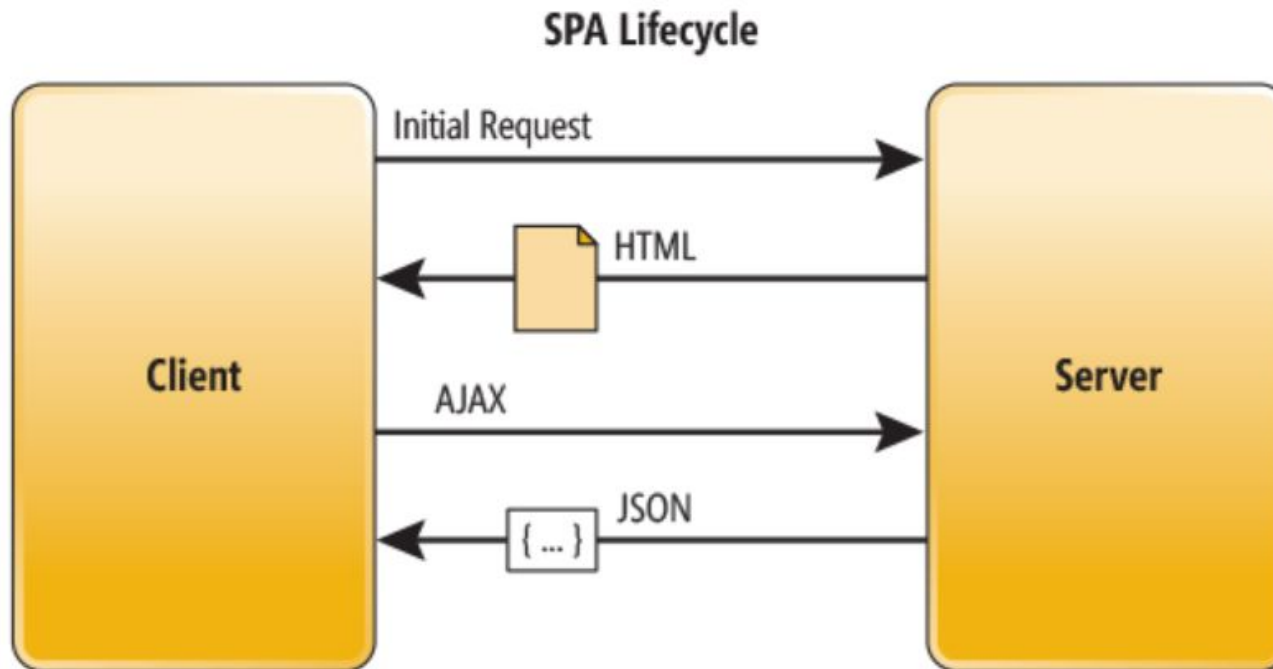
Multi-Page Applications

1. Client-Side Routing
2. Multi-Page Applications



Single Page Applications

1. Client-Side Routing
2. Multi-Page Applications
3. Single-Page Applications



React Router

1. **Client-Side Routing**
2. **Routing?**
 - **Directing to Different Pages**
3. **React Router**

Need for React Router

1. Navigation Across Multiple Pages

Enables navigation between different views within a React application.

2. React Router

- Routing library for React
- Client Sides Routing

3. SPAs without Page Refresh

4. Allows Utilization of Browser History Feature

Utilizes the browser's History API

5. Provides Routing Components

BrowserRouter, Routes, Route, Link, NavLink

React Router - Packages

1. **react-router**
2. **react-router-dom**
3. **react-router-native**

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1. **react-router**
 1. **Core package**
 2. **Components and Functions**
2. **react-router-dom**
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 1. **Mobile Applications**

React Router - Packages

1. **react-router**
 1. **Core package**
 2. **Components and Functions**
2. **react-router-dom**
 1. **Browser**
 2. **npm install react-router-dom@latest**
3. **react-router-native**
 1. **Mobile Applications**

React Router – Simple Routing

Home
Contact
About

Welcome to my Homepage.

React Router Steps

- 1. React Router Installation**
- 2. Creating Necessary React Pages/Components (Separate jsx files)**
- 3. Import Target React Pages/Components (App.jsx)**
- 4. Importing Necessary Packages from react-router-dom (App.jsx)**
- 5. Add BrowserRouter and Routes (App.jsx)**
- 6. Add Corresponding Route (App.jsx)**
- 7. Link to Navigate to Components (App.jsx)**

Simple Routing

1. React Router Installation

- **`npm install react-router-dom@latest`**

Simple Routing

Step 1 — Go to your working folder

Open the terminal and run:

```
cd /home/rourab/WebTech_lab
```

Simple Routing

Step 2 — Create a new React project

Run:

```
npm create vite@latest react-routing-example -- --template react
```

It will ask some questions. If prompted, choose **React** and **JavaScript**.

Then enter the project:

```
cd react-routing-example
```

Simple Routing

Step 3 — Install the dependencies

Run:

```
npm install
```

Then install React Router:

```
npm install react-router-dom@latest
```

Simple Routing

Step 4 — Start the project once

Run:

```
npm run dev
```

You should see something similar to:

Local: <http://localhost:5173/>

Open that URL.

You will initially see the default Vite React page. That's normal.

Simple Routing

Step 5 — Create our routing structure

[sample code](#)

Now your project should look like:

```
react-routing-example/  
├── package.json  
├── index.html  
├── vite.config.js  
└── src/  
    ├── App.jsx  
    ├── main.jsx  
    ├── App.css  
    ├── index.css  
    └── assets/
```

We will change it to:

```
react-routing-example/  
├── package.json  
├── index.html  
├── vite.config.js  
└── src/  
    ├── App.jsx  
    ├── main.jsx  
    ├── style.css  
    └── pages/  
        ├── Home.jsx  
        ├── About.jsx  
        └── Contact.jsx
```

Simple Routing

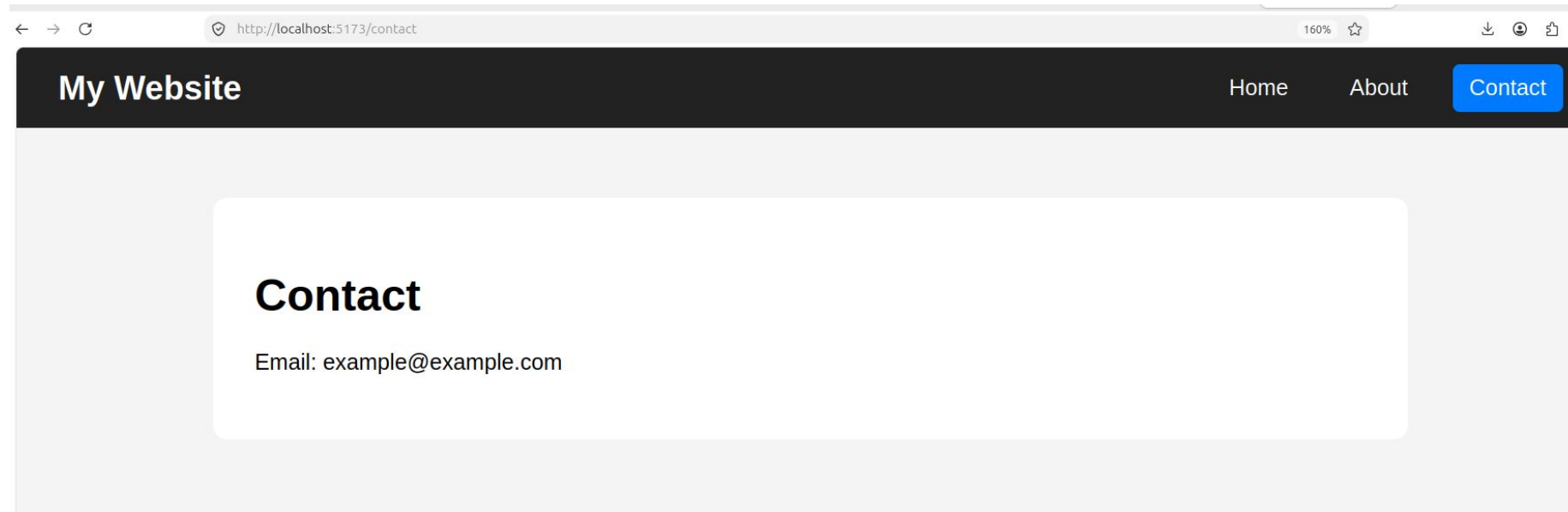
Step 6 — Download sample code

paste the pages inside src directory

replace App.jsx, main.jsx and style.css in src directory

run

npm run dev



Advantages of React Router

- 1. Navigation without full page reload**
- 2. Supports Single Page Applications**
- 3. Clean and meaningful URLs**
- 4. Browser Back/Forward support**
- 5. Component-based and dynamic routing**

References

1. **Robin Wieruch, “The Road to React: The ReactJS with Hooks”, Lean Publishing, 2023 Edition.**

THANK YOU

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